

Graduate-Level Qualifying Exam Problems on Slutsky's Theorem

Ten proof-oriented problems with hints only. No solutions are provided.

Problem 1. Prove the addition form of Slutsky's theorem directly from the definition of convergence in probability, without invoking the continuous mapping theorem.

Hint: Use an $\epsilon/2$ argument and the union bound.

Problem 2. Prove the multiplication form of Slutsky's theorem: if $X_n \rightarrow X$ in probability and $Y_n \rightarrow c$ in probability, then $X_n Y_n \rightarrow cX$ in probability.

Hint: Write $X_n Y_n - cX = X_n(Y_n - c) + c(X_n - X)$ and control X_n .

Problem 3. Prove the division form of Slutsky's theorem under the assumption $c \neq 0$.

Hint: First show that Y_n stays away from zero with high probability.

Problem 4. Let $X_n \rightarrow X$ in distribution and $Y_n \rightarrow c$ in probability. Prove that $X_n + Y_n \rightarrow X + c$ in distribution.

Hint: Apply a distribution-function argument or Portmanteau theorem.

Problem 5. Show that if $X_n \rightarrow X$ in distribution and $Y_n \rightarrow c$ in probability, then $X_n Y_n \rightarrow cX$ in distribution.

Hint: Reduce to the addition case using continuous transformations.

Problem 6. Suppose $\sqrt{n}(X_n - \theta) \Rightarrow Z$ and $Y_n \rightarrow a \neq 0$ in probability. Show that $\sqrt{n}(X_n Y_n - a\theta) \Rightarrow aZ$.

Hint: Expand around $a\theta$ and use Slutsky.

Problem 7. Let X_n be asymptotically normal with mean μ and variance σ^2/n . Show that $X_n/(1+X_n)$ has an asymptotic distribution and identify it.

Hint: Use the delta method after establishing consistency.

Problem 8. Suppose $X_n \Rightarrow X$ and $Y_n \Rightarrow Y$ jointly, where Y is degenerate at c . Prove Slutsky's theorem using the continuous mapping theorem.

Hint: Form the vector (X_n, Y_n) .

Problem 9. Let T_n be a consistent estimator of θ and S_n be a consistent estimator of σ^2 . Show that the Studentized statistic $(T_n - \theta)/\sqrt{S_n}$ has the same asymptotic law as $(T_n - \theta)/\sigma$ whenever an asymptotic distribution exists.

Hint: Use Slutsky with $\sqrt{S_n} \rightarrow \sigma$.

Problem 10. Give a complete proof that the class of statistics obtained from consistent estimators by finitely many additions, multiplications, and divisions by nonzero consistent estimators is itself consistent.

Hint: Proceed by induction using the various forms of Slutsky's theorem.